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Electric Light in the Lecture Hall

Last evening a compact arc lamp was exhibited before the members of the Mechanics' Institute. Two carbon rods were advanced by a clockwork movement, while the current was supplied by a small dynamo in the adjoining room. The illumination was strong enough to project a spectrum upon a screen at the far end of the hall.

The operator first showed the unequal consumption of the carbons, then reversed the current so that the crater appeared upon the opposite rod. A white card, held at several distances, made the rapid fall of brightness apparent without requiring a photometer.

"A measurement is most useful when its conditions travel with it."

The demonstration was notable less for novelty than for order. Every change of distance, polarity, and aperture was announced before the observation was made. The record may therefore be repeated instead of merely admired.

Notes and Correspondence

A correspondent at Bath reports that a magnetic needle, suspended beneath a glass bell, retained its period after the air had been partly withdrawn. Temperature was not recorded, and the observation should be treated as provisional.

Books received.—A new table of logarithms; a practical treatise on the construction of lenses; and the second part of a catalogue of British mosses.

The Transit Instrument

The observatory has received a transit telescope with a divided circle of twelve inches diameter. Its axis rests upon stone piers separated from the surrounding floor, so that the motion of visitors is not readily communicated to the instrument.

Before the first observations, the collimation, level, and azimuth errors were determined separately. Repeating the same star upon both faces of the circle then exposed an index error that a single reading would have concealed.

Meteorological Register

Day	Bar.	Temp.	Rain
Monday	29.84	54.2	0.00
Tuesday	29.71	51.8	0.13
Wednesday	29.92	56.1	0.02
Thursday	30.06	58.7	0.00
Friday	30.11	60.3	0.00

Barometer reduced to 32 degrees Fahrenheit; rain in inches. The ruled column divider, masthead, compact heads, and tabular register are independent newspaper mechanisms rather than a single background texture.